Malaysia Energy Transition

- Overview
- Generation
- Regions
- State reliability
- Grid load
- Forecast
- ASEAN
- About
- Data quality

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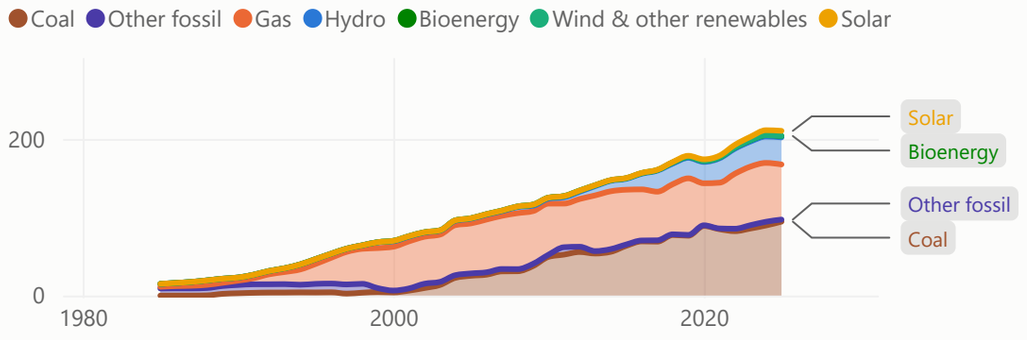
Renewables supplied 20.4% of Malaysia's electricity in 2025

Malaysia · 2025 · Data as of 10 Oct 2026

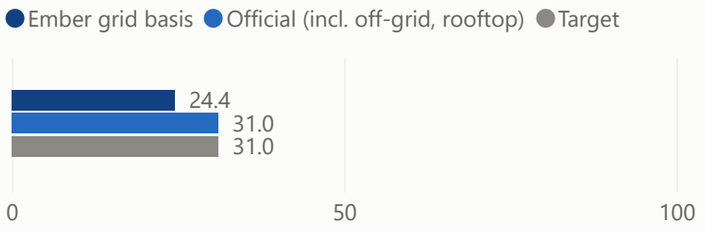
Year: All

210.4	20.4	79.6	606	5.81
Total Generation (TWh)	RE Share (%)	Fossil Share (%)	Carbon Intensity (gCO2e/kWh)	Demand per Capita (MWh)
▼ 0.4% vs 2024	▲ 0.6 pp vs 2024	▼ 0.6 pp vs 2024	▲ 0.2% vs 2024	▼ 1.6% vs 2024

Malaysia's generation mix by source (TWh), all years



Renewable share of installed capacity (%)



Ember grid basis 24.4% · official 31% (incl. off-grid and rooftop, 2025) · target 31% by 2025

Malaysia ranks 4 of 6 on renewable share (2025) →

Headline findings (2015–2025, Ember basis; caveats in docs/findings.md)

1. The renewable share roughly doubled (9.6% → 19.7%, 3-year averages), mostly in a 2015–17 hydro step; since 2017 hydro and solar share the growth.
2. Each kWh got 9.5% cleaner, yet power-sector emissions rose 27%: generation grew 40%, and all the fossil growth was coal.
3. Malaysia is 4th of 6 ASEAN peers on renewable share and 5th of 6 on carbon intensity (606 g/kWh, 2025): the coal-to-gas balance matters as much.

Sources: Ember Yearly Electricity Data (CC BY 4.0); OWID population (CC BY 4.0). Targets: 31% by 2025 and 40% by 2035 (12MP / SEDA MyRER, 2021), 35% by 2030 (NETR pathway), 70% by 2050 (NETR, 2023). Ember counts grid capacity only; the official share also counts off-grid and rooftop capacity. Large hydro is renewable in both. KPIs describe Malaysia in the selected year.

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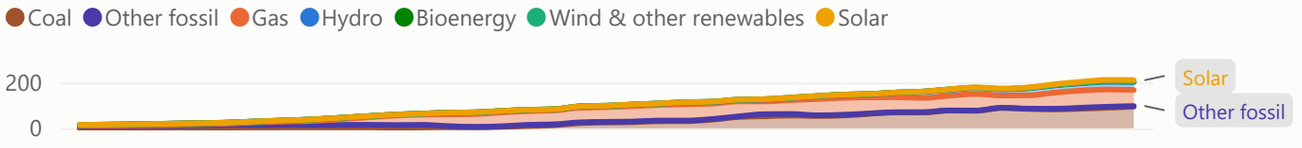
↺ Reset filters

Malaysia: fossil fuels supplied 79.6% of electricity in 2025

Malaysia · 1985–2025 · Data as of 10 Oct 2026

Years: 1985 – 2025

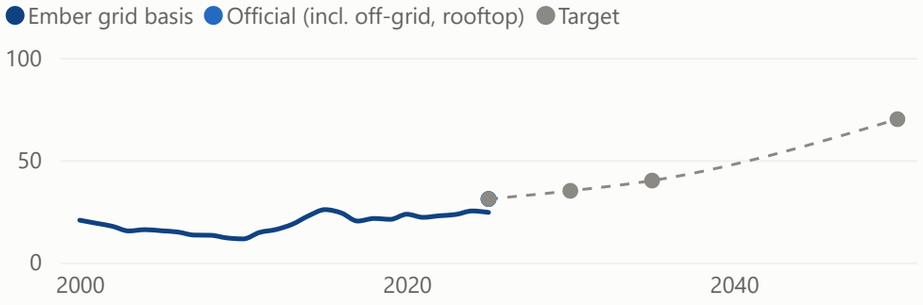
Generation by source (TWh or % of the year's total, per the Unit toggle)



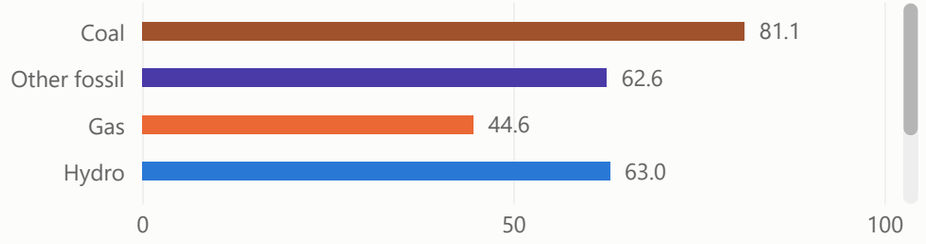
Unit: TWh | %

Source: All

National RE capacity targets: share of installed capacity (%)



Capacity factor by source, 2025 (%)



Electricity demand (TWh)



Demand growth, year on year (%)



Sources: Ember Yearly Electricity Data (CC BY 4.0). Capacity factor = generation ÷ (capacity × hours in the year; 8,784 h in leap years), shown at the last year of the selected range; blank, never 0, when capacity is not reported. Targets: 31% by 2025 and 40% by 2035 (12MP / SEDA MyRER, 2021); 35% by 2030 and 70% by 2050 (NETR). Ember counts grid capacity only; the official share also counts off-grid and rooftop capacity. The target chart always spans 2000-2050 and ignores the year range.

A Sabah customer lost supply about 13× as often as a Peninsular one in 2022

Peninsular Malaysia, Sabah (incl. Labuan) and Sarawak · capacity, use and supply reliability · Source: Energy Commission

Peninsular Malaysia

Capacity MW at 31 Dec 2021 · use 2021 · SAIFI 2022

Installed	Available	Use GWh
29,410	26,491	118,365
kWh/head	SAIFI	
4,449	0.84	

Sabah

Capacity MW at 31 Dec 2021 · use 2021 · SAIFI 2022

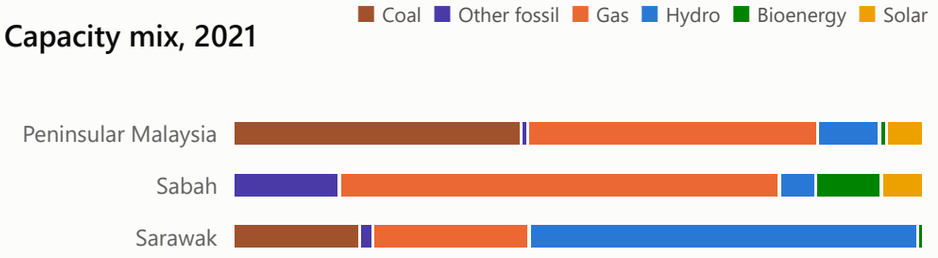
Installed	Available	Use GWh
1,866	1,314	5,894
kWh/head	SAIFI	
1,680	11.07	

Sarawak

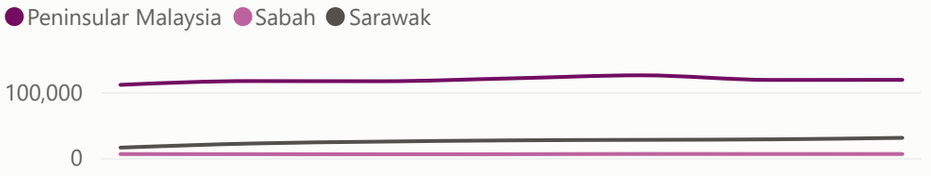
Capacity MW at 31 Dec 2021 · use 2021 · SAIFI 2021

Installed	Available	Use GWh
6,147	5,439	30,446
kWh/head	SAIFI	
12,348	1.01	

Capacity mix, 2021



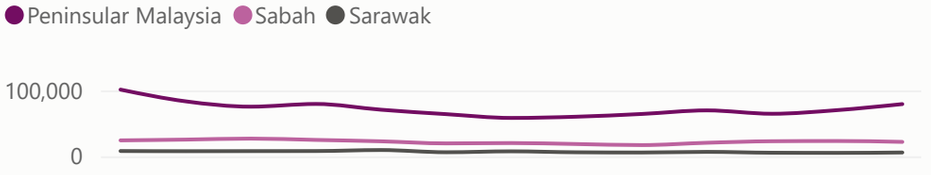
Electricity consumption by region (GWh)



SAIFI by region: Sabah's is 13.2× Peninsular's



Supply interruptions by region (number per year)



Sarawak's reliability table prints its years one year early; years are mapped from the 2022 edition (2015-2021), and its unproven last column is left out. Peninsular SAIFI/SAIDI are the state tables' totals.

Source: Energy Commission, Malaysia Energy Statistics Handbook 2023 (pp. 13-19, 82-85). Charts adapted from the handbook with the Energy Commission's written permission (9 October 2026). Not endorsed by the Energy Commission. Sabah includes Labuan.

Kelantan had the least reliable supply in 2022, Putrajaya the most (SAIFI)

Peninsular Malaysia · SAIFI, interruptions per customer per year · 2022 · Source: Energy Commission

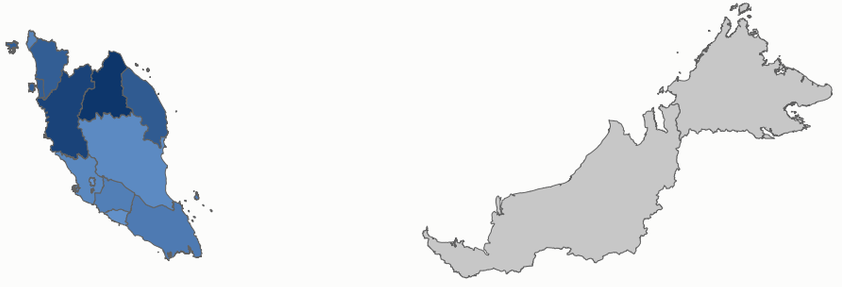
SAIFI | SAIDI

Year

All

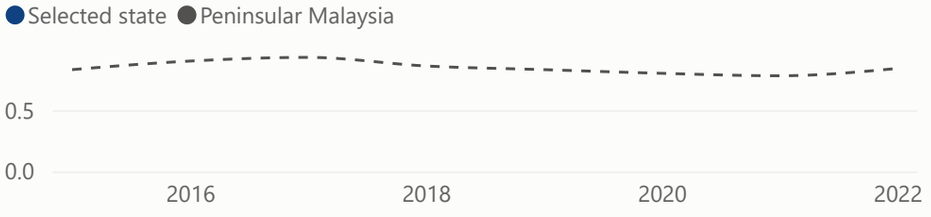
Year: leave empty for the latest (2022). CAIDI (not shown) is misaligned for four states in 2015-2018 in both the 2022 and 2023 editions; SAIDI and SAIFI are shown as published.

SAIFI by state, 2022 (darker = less reliable)

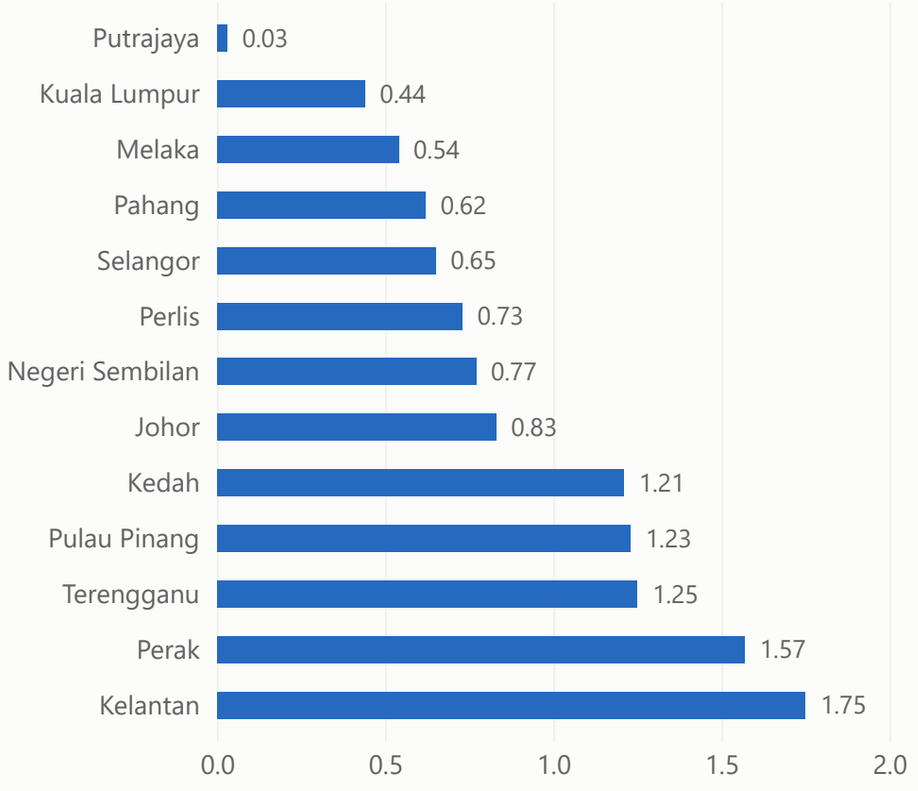


Grey: Sabah, Sarawak and Labuan are not reported by state (see Regions).

Peninsular Malaysia (click a state to compare) · SAIFI by year



SAIFI by state, 2022 (rank 1 = most reliable)



Source: Energy Commission, Malaysia Energy Statistics Handbook 2023 (pp. 83-84). Charts adapted from the handbook with the Energy Commission's written permission (9 October 2026). Not endorsed by the Energy Commission. Boundaries © OpenStreetMap contributors, via geoBoundaries (ODbL 1.0). Putrajaya includes Cyberjaya.

Thailand: demand peaked at 35,560 MW in 2024; load factor 69%

Sum of regional demand (EGAT) · hourly, 2023–24 · local time · Data as of 10 Oct 2026

Grid: Thailand national grid (EGAT)

Showing: Thailand national grid (EGAT) · Open-data stand-in for Peninsular Malaysia until TNB consents to GSO data use

Average load by month and hour (GW)

Peak 35,560 MW on 29 Apr 2024, 21:00

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Jan	20.2	19.4	18.8	18.2	18.0	18.2	19.5	19.8	19.5	20.4	20.3	20.3	19.0	19.6	21.3	21.8	22.4	21.9	23.0	24.7	23.9	23.2	22.0	21.8
Feb	22.9	21.9	21.3	20.5	20.1	20.3	21.5	21.4	21.3	22.4	22.5	22.5	21.3	21.9	23.7	24.3	24.7	23.9	24.3	26.8	26.3	26.0	24.9	24.6
Mar	25.2	24.1	23.3	22.5	22.0	22.0	22.8	22.2	22.6	23.9	24.1	24.3	23.2	24.0	26.0	26.5	26.8	25.7	25.8	28.3	28.3	28.2	27.3	27.0
Apr	27.7	26.5	25.5	24.4	23.7	23.5	23.7	22.4	23.2	24.6	25.1	25.5	24.8	25.7	27.5	28.0	28.2	26.7	26.5	29.2	29.9	30.4	29.8	29.4
May	27.4	26.2	25.3	24.3	23.7	23.5	23.7	22.5	23.3	25.0	25.6	26.0	25.1	26.0	27.9	28.2	28.1	26.7	26.6	29.2	29.9	30.3	29.5	29.0
Jun	26.1	25.0	24.1	23.2	22.7	22.7	23.0	21.8	22.5	24.2	24.6	24.9	23.9	24.7	26.4	26.6	26.6	25.3	25.5	28.0	28.8	29.0	28.1	27.7
Jul	24.7	23.7	23.0	22.2	21.7	21.8	22.5	21.5	22.0	23.5	23.9	24.0	23.0	23.6	25.2	25.5	25.5	24.4	24.6	27.0	27.6	27.6	26.6	26.3
Aug	24.9	23.9	23.1	22.3	21.8	21.9	22.7	21.6	22.1	23.6	24.0	24.3	23.3	24.0	25.8	26.1	26.2	25.1	25.4	28.0	28.2	28.1	27.0	26.6
Sep	24.1	23.2	22.5	21.7	21.3	21.5	22.5	21.5	21.9	23.4	23.7	23.9	22.9	23.5	25.2	25.6	25.8	24.8	25.6	27.8	27.5	27.1	25.9	25.6
Oct	23.9	23.0	22.3	21.6	21.2	21.3	22.1	21.2	21.7	22.9	23.3	23.5	22.6	23.3	25.1	25.5	25.8	24.8	26.5	27.6	27.2	26.7	25.6	25.4
Nov	23.1	22.3	21.6	20.9	20.6	20.8	22.0	21.2	21.1	22.2	22.3	22.6	21.6	22.3	24.2	24.8	25.3	24.6	26.8	27.2	26.5	26.0	24.8	24.6
Dec	21.1	20.3	19.7	19.0	18.7	19.0	20.1	19.8	19.6	20.2	20.2	20.1	19.1	19.7	21.4	22.1	22.8	22.6	24.6	25.3	24.6	23.9	22.8	22.3

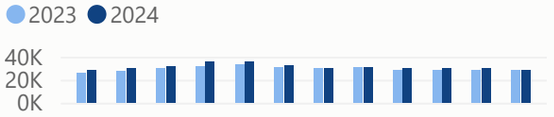
Typical day (MW)



Season

- [ ] Hot (Mar–May)
- [ ] Rainy (Jun–Oct)
- [ ] Cool (Nov–Feb)

Monthly peak (MW)



Load factor (%)



Thailand ↔ Malaysia daily net flow (MW)

Exported 235.3 GWh to Malaysia and imported 65.5 GWh in 2023; 149.6 and 150.9 GWh in 2024. Above 0 = Thailand exports



Thailand: sum of 5 regional demands, hourly means (published peaks are 2–4% higher). Singapore: EMA system demand, hourly mean of half-hours. Seasons are month-based. Gaps shown, never filled. Flows: Thailand's malaysia_EXP/IMP, daily means. Data: EGAT via Zenodo (CC BY 4.0); EMA (SODL v1.0).

Thailand: day-ahead forecast cuts hourly error by 47%

Sum of regional demand (EGAT) · day-ahead backtest, 2024 · local time · Data as of 10 Oct 2026

Grid: Thailand national grid (EGAT)

Showing: Thailand national grid (EGAT) · Open-data stand-in for Peninsular Malaysia until TNB consents to GSO data use

Backtest day

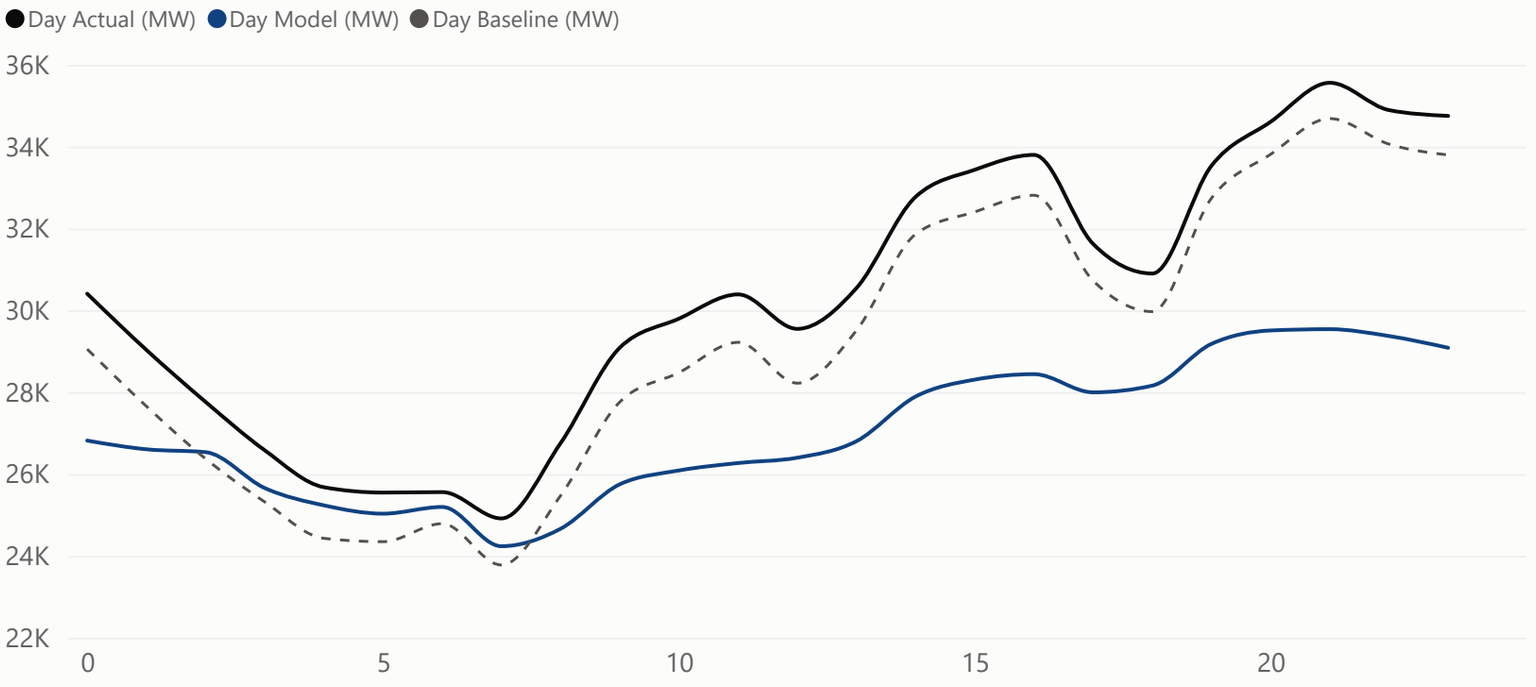
All

Beats the seasonal-naive baseline: 46.9% lower MAPE (95% CI 39.3 to 52.8)

Also beats a weekday-aware naive rule: 33.9% lower MAPE (95% CI 26.0 to 40.1)

Backtest 1 Jan 2024 – 31 Dec 2024 · 8,778 hours scored, 6 excluded

Forecast vs actual, Mon 29 Apr 2024 (2024 peak day)



3.31	6.23	46.9	810	1,485
Model MAPE (%)	Baseline MAPE (%)	Improvement (%)	Model MAE (MW)	Baseline MAE (MW)

Forecast: LightGBM issued at local midnight for the next 24 h, monthly rolling-origin backtest over 2024, run once under a locked protocol (docs/forecast.md). Baseline = the same hour last week; improvement = 1 − MAPE model ÷ MAPE baseline, 95% CI from a 7-day block bootstrap. Pick a day; none picked = the grid's 2024 peak day.

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Malaysia: 4 of 6 on renewable share, 5 of 6 on carbon intensity (2025)

ASEAN peers · 1985–2025 · Data as of 10 Oct 2026

Years: 1985 – 2025

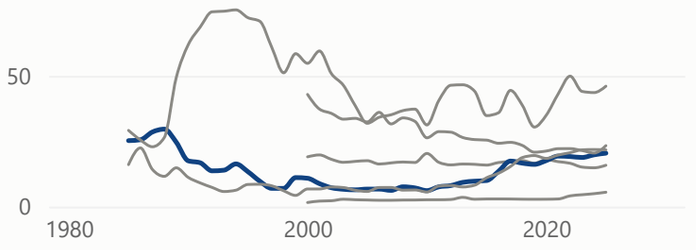
Countries (none ticked = 6...

- [ ] Malaysia
- [ ] Indonesia
- [ ] Thailand
- [ ] Vietnam
- [ ] Philippines
- [ ] Singapore
- [ ] Brunei
- [ ] Cambodia
- [ ] Laos

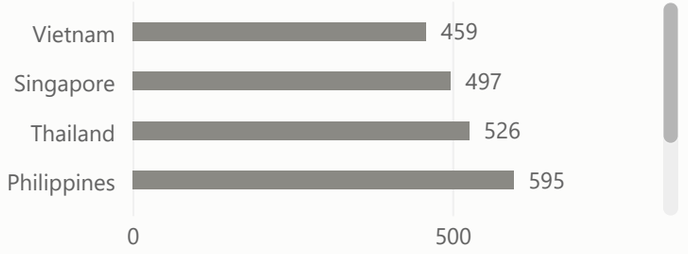
Trend metric

- [ ] RE share (%)
- [ ] Carbon intensity (gCO2e/kWh)
- [ ] Emissions (MtCO2e)

Trend by country (Malaysia in navy; pick the metric on the left)



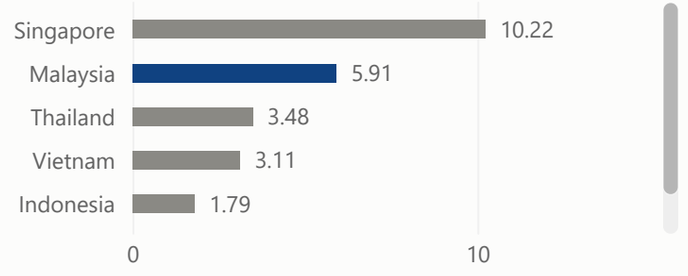
Carbon intensity at the comparison year (gCO2e/kWh), lowest first



RE share vs carbon intensity (bubble = demand, TWh)



Demand per capita at the common year (MWh)



Malaysia ranks 4 of 6 on renewable share (2025)

Ranking, scatter and intensity bars use 2025, the latest year all selected countries report; per-capita demand uses 2024 (OWID population). With no country selected, the six default peers are compared. Sources: Ember (CC BY 4.0), OWID (CC BY 4.0).

Where the data comes from, and on what terms

Run 47 · WARN · started 2026-10-10 13:04, finished 2026-10-10 13:07 UTC

Sources and licences

Source	What it provides	Licence	Credit	Consent	Link
Department of Statistics Malaysia via data.gov.my	Malaysia electricity supply, a cross-check only (series ends June 2024)	CC BY 4.0	Department of Statistics Malaysia, data.gov.my, CC BY 4.0	OPEN	
Ember Yearly Electricity Data	Annual generation, capacity and emissions by source for the ASEAN countries	CC BY 4.0	Ember (2026), Yearly Electricity Data, CC BY 4.0	OPEN	
Energy Commission Malaysia Energy Statistics Handbook	Regional capacity, use and state reliability (SAIFI/SAIDI); derived charts only	Energy Commission written consent (granted 2026-10-09)	Source: Energy Commission, Malaysia Energy Statistics Handbook 2023 (charts adapted)	GRANTED	
Energy Market Authority Singapore half-hourly system demand	Singapore half-hourly system demand (grid load and forecast pages)	Singapore Open Data Licence v1.0	Contains information from Half-hourly System Demand Data accessed on 2026-10-03 from the Energy Market Authority of Singapore, made available under the terms of the Singapore Open Data Licence version 1.0 (https://www.ema.gov.sg/terms-of-use)	OPEN	

KPI definitions

KPI	Definition	Formula	Unit
Capacity factor	How much a source generated compared with running at full capacity all year.	Generation (TWh) × 1000 ÷ (capacity (GW) × hours in year; 8784 in leap years)	%
Carbon intensity of electricity	The greenhouse-gas emissions released per unit of electricity generated in a year.	Power-sector emissions ÷ total generation (Ember 'Emissions intensity')	gCO_2e/kWh
Demand growth	Year-on-year change in a	(Demand this year −	%

Method notes

Forecast: LightGBM issued at local midnight, scored once on a locked 2024 backtest against a seasonal-naive baseline (the same hour last week). The verdict is on the Forecast page.
Capacity factor = generation ÷ (capacity × hours in the year); blank if capacity is unreported.
States: handbook labels map to states through a reference table; reliability by state covers Peninsular states only, and Sabah includes Labuan.
Sarawak: the 2023 handbook prints each Sarawak reliability year one year early, so the years are remapped (see docs/reconciliation.md).
Data lag: Ember runs to 2025 (Brunei, Laos and Myanmar to 2024); grid load covers 2023-24; DOSM supply ends in June 2024 and is a cross-check only.

Energy Commission Malaysia Energy Statistics Handbook: permission granted; Grid System Operator (GSO) System Demand: permission requested

Source code on GitHub

Credits are worded as each licence asks. Boundaries © OpenStreetMap contributors, via geoBoundaries (ODbL 1.0). A source marked REQUESTED needs written permission and is used locally only until it is granted. Links open in your browser (Ctrl+click in Power BI Desktop). Licence texts: DATA_LICENSES.md in the repository.

Every data-quality rule, checked on each pipeline run

Results of the most recent run · pass, warn or fail · click a source to see its rules

Run 47 · WARN · started 2026-10-10 13:04, finished 2026-10-10 13:07 UTC

30 pass · 4 warn · 0 fail, of 34 rules

PASS | WARN

Rows per source (click one to filter the rules)

Source	Rows read	Rows changed	Last loaded
Department of Statistics Malaysia via data.gov.my	1,224	599	2026-10-10
Ember Yearly Electricity Data	5,070	0	2026-10-10
Energy Commission Malaysia Energy Statistics Handbook	989	0	2026-10-10
Energy Market Authority Singapore half-hourly system demand	106,848	0	2026-10-10

Expected WARNs (docs/reconciliation.md)

OWID_GEN_WITHIN_5PCT: OWID uses a different upstream vintage than Ember.
DOSM_SUPPLY_WITHIN_10PCT: DOSM's supply is a net balance below Ember's gross generation, and Ember revised 2021 on.
CF_AT_MOST_1: Ember's capacity is too small for the generation it

Data-quality rules

Rule	Source	Severity	Result	Rows affected
CF_AT_MOST_1	D1	WARN	! Warn	117
STALE_COUNTRY_YEARS	D1	WARN	! Warn	3
OWID_GEN_WITHIN_5PCT	D2	WARN	! Warn	2
DOSM_SUPPLY_WITHIN_10PCT	D3	WARN	! Warn	1
CAIDI_CONSISTENT	D5	WARN	✓ Pass	0
CAPACITY_MATCHES_TOTALS	D5	BLOCKING	✓ Pass	0
CAPACITY_SUBTOTALS_SUM	D5	BLOCKING	✓ Pass	0
CAPACITY_WITHOUT_GENERATION	D1	WARN	✓ Pass	0
DEMAND_PLAUSIBLE		WARN	✓ Pass	0
FLOW_ONE_DIRECTION		WARN	✓ Pass	0
GRID_TIMESTAMP_PARSED		BLOCKING	✓ Pass	0
HANDBOOK_PEN_TOTALS_AGREE	D5	WARN	✓ Pass	0
HANDBOOK_RAW_SHAPE	D5	BLOCKING	✓ Pass	0
HANDBOOK_VALUES_CAST	D5	BLOCKING	✓ Pass	0
HOURLY_GAPS_LE_2PCT		BLOCKING	✓ Pass	0
INTENSITY_RECOMPUTED	D1	WARN	✓ Pass	0
MIX_SUMS_TO_100	D1	BLOCKING	✓ Pass	0
MIX_TOTAL_MATCHES	D1	WARN	✓ Pass	0
NO_DUPLICATE_STATE_YEAR	D5	BLOCKING	✓ Pass	0

Rules run on every pipeline run; the number on the Data quality button counts rules that FAIL. The four WARNs above are expected differences between sources, not pipeline faults (docs/reconciliation.md). Click a source on the left to see only its rules; click it again to clear. Rules with no source check the grid data.